MALEK BENCHEHIDA

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EDUCATION

University of Illinois Chicago (UIC) Chicago, IL

Bachelor of Science in Computer Science **Expected Graduation: December 2026**

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Computer Algorithms, Software Design, Distributed Systems, Computer Systems, Discrete Mathematics.

TECHNICAL SKILLS

- **Languages:** C++ (C++17/20, STL, Multi-threading), Python, Rust, C, Go, SQL, TypeScript
- **Robotics & Planning Concepts:** Multi-Agent Pathfinding (MAPF), Heuristic Search (A^*, Dijkstra, CBS), Graph Search, Motion Planning, State Machines, Collision Avoidance, Task Allocation, Optimization
- **Systems & Concurrency:** Real-Time Control Loops, Concurrency & Mutexes, Asynchronous I/O, Distributed Systems, Event-Driven Architecture, Microservices, RTOS, POSIX Linux
- **Hardware & Protocols:** NVIDIA Jetson AGX Thor, CAN Bus, I2C, SPI, UART, Microcontrollers (STM32, ESP32)
- **Developer Tools:** Linux/Unix, Git, Docker, CMake, GDB, Valgrind, CI/CD, AWS, Azure, Telemetry & Profiling

ROBOTICS & EMBEDDED SYSTEMS EXPERIENCE

SAE Formula Electric Team — UIC Chicago, IL

Technical Software Lead / Embedded Systems Engineer **2024 – Present**

- Architect real-time embedded control loops in **C++** for deterministic powertrain actuation, throttle-pedal monitoring, and closed-loop motor velocity control under microsecond latency constraints.
- Design and implement a robust telemetry and diagnostic pipeline over the **CAN Bus** protocol, coordinating inter-ECU message arbitration and priority queues across distributed vehicle sensor nodes.
- Refactor safety-critical control modules from C++ into **Rust**, enforcing compile-time memory safety, thread safety, and zero-cost abstractions to eliminate race conditions in concurrent data paths.
- Lead an engineering sub-team in continuous integration, unit testing, and hardware-in-the-loop (HIL) validation, benchmarking firmware stability against edge-case vehicle dynamics.

EdgeGuardNano — Edge Computing & Autonomous Telemetry **2026**

Systems & ML Developer

- Engineered an edge-compute autonomous telemetry and inference system deployed directly onto an **NVIDIA Jetson AGX Thor** platform for real-time, offline environment monitoring.
- Built asynchronous, non-blocking ingestion pipelines in **Rust** and **C++**, maximizing ARM64 CPU throughput and ensuring deterministic execution during continuous high-rate sensor streams.
- Profiled memory hierarchy, cache utilization, and thread scheduling to eliminate processing bottlenecks, achieving low-latency processing and reliable edge-node classification under resource constraints.

Reaction-Time Distributed Control System **2026**

Embedded Systems Developer

- Designed a distributed multi-microcontroller architecture utilizing **I2C** communication to arbitrate real-time concurrent input events across master-slave nodes with sub-millisecond response accuracy.
- Implemented deterministic state machine logic in C++ to handle packet collision resolution, state synchronization, and fault-tolerant event scoring under rapid continuous input bursts.

SOFTWARE & SYSTEMS EXPERIENCE

MatchNice Chicago, IL (Remote)

Software Engineer **June 2026 – Present**

- Design high-throughput event processing pipelines and distributed background sync workers, enforcing concurrency locks and idempotency to guarantee 100% state consistency across asynchronous transactional streams.
- Architect scalable microservice APIs and data workflows handling concurrent external events with automated retry strategies, backoff mechanisms, and validation schemas.

CAS Enterprise Inc. Chicago, IL

DevOps & Systems Engineer **Aug 2023 – June 2026**

- Monitored system bottlenecks and optimized backend data pipelines, diagnosing memory leaks and tracing unindexed queries to reduce latency and eliminate operational downtime.
- Automated distributed build, test, and deployment workflows using **Python** and CI/CD pipelines, accelerating delivery cycles by 91% while maintaining production fault tolerance.